

Multiple Choice (10 marks)

1. Man in the middle attack can endanger the security of Diffie Hellman method if two parties are not
 - (a) Joined
 - (b) Authenticated
 - (c) Submitted
 - (d) Shared
2. Which of the following is not a security exploit?
 - (a) Eavesdropping
 - (b) Cross-site scripting
 - (c) Authentication
 - (d) SQL Injection
3. Encapsulating Security Payload (ESP) belongs to which Internet Security Protocol?
 - (a) Secure Socket Layer Protocol
 - (b) Secure IP Protocol
 - (c) Secure Http Protocol
 - (d) Transport Layer Security Protocol
4. Which of the following pairs of 8-bit, two's-complement numbers will result in overflow when the members of the pairs are added?
 - (a) 11111111, 00000001
 - (b) 00000001, 10000000
 - (c) 11111111, 10000001
 - (d) 10000001, 10101010
5. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?
 - (a) To translate Web addresses to host names
 - (b) To determine the IP address of a given host name
 - (c) To determine the hardware address of a given host name
 - (d) To determine the hardware address of a given IP address

True / False (10 marks)

Answer True or False

6. True or False: Whitebox fuzzers are designed to explore all possible execution paths in a program, ensuring comprehensive code coverage.
7. True or False: Poor handling of unexpected input is a common example of presentation layer issues in software design.
8. True or False: Wireless security is the practice of preventing unauthorized access to computers or data through wireless networks.
9. True or False: A stream cipher can achieve perfect secrecy if the pseudorandom generator is sufficiently secure.
10. True or False: A buffer overflow occurs when writing to a pointer that has been freed, leading to data corruption.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check whether the given month number contains 28 days or not.
12. Write a function to get the word with most number of occurrences in the given strings list.

End of Paper